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THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF ELECTRICAL ENGINEERING AND
TELECOMMUNICATIONS

ELEC4631
Continuous-Time Control System Design

Final Examination

T2 2021

- Time Allowed: 2 hours 5 minutes (exam time) plus 30 minutes (scanning and uploading solutions)
- This examination is open-book. You may use course notes and other course materials.
- NO collaborations allowed and the work submitted must be your own. During the exam you may NOT be in contact with anyone else via any form of communication media (email, messaging, phone, video conferencing, etc). Violating this will be considered an academic misconduct and disciplinary action will be taken against anyone who is proven to have violated this rule.
- This paper contains this page and 4 other pages containing four (4) Questions. There is a total of 5 pages including this page.
- Total marks available: 100.
- This examination is worth 56% of the total course mark.
- All questions are NOT of EQUAL value. Answer ALL questions.
- You may use a calculator or computer software such as MATLAB only for real and complex arithmetic, evaluating powers (squares, cubes, etc), roots (square roots, cube roots, etc) and the standard functions e^x , 10^x , $\ln(x)$, $\log_{10}(x)$, $\sin(x)$, $\cos(x)$ and $\tan(x)$ for real numbers x , and for the numerical addition and multiplication of matrices. You may not use Matlab or any other software for symbolic manipulations. ALL workings must be shown.
- All answers must be written in ink. Except where they are expressly required, pencils may be used only for drawing, sketching or graphical work.
- Uploaded scans must be readable, clear and of appropriate size. That is, they must be fit for marking. Questions may not be marked and will receive zero marks if they are not legible.
- Scan and upload your solutions to all problems together in a single pdf file.

QUESTION 1

(20 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Consider the SISO LTI state-space system

$$S = \left(\begin{bmatrix} -n_4 & n_5 + 2 \\ 1 & -n_6 \end{bmatrix}, \begin{bmatrix} n_1 + 1 \\ 2n_1 + 2 \end{bmatrix}, \begin{bmatrix} 0 & n_2 + 2 \end{bmatrix}, 0 \right).$$

Answer the following questions:

- (a) Determine the internal stability property that the system has (asymptotically stable, marginally stable, or unstable).

(5 marks)

- (b) Compute the system's transfer function.

(4 marks)

- (c) Find a transformation T such that $S' = (TAT^{-1}, TB, CT^{-1}, 0)$ is in observer canonical form.

(5 marks)

- (d) Consider the LTI state-space system

$$S = \left(\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}, \begin{bmatrix} 8 \\ 16 \end{bmatrix}, \begin{bmatrix} -1 & 1 \end{bmatrix}, 0 \right).$$

The above system is neither controllable nor observable. Without computing the controllability and observability matrices (since you have been informed that none of them are invertible), explain why this system fails both the definitions of a controllable system and an observable system. **Hint:** No lengthy and complicated calculations required. An appropriate similarity transformation may be useful.

(6 marks)

⁰Please turn over - Question 2 on next page.

QUESTION 2

(30 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Consider the SISO LTI state-space system,

$$S = \left(\begin{bmatrix} 0 & -n_5 \\ 1 & -n_6 \end{bmatrix}, \begin{bmatrix} n_1 + 1 \\ 2n_1 + 2 \end{bmatrix}, \begin{bmatrix} 0 & n_2 + 2 \end{bmatrix}, 0 \right).$$

Answer the following questions:

- (a) Suppose that you have access to the state $x(t)$. Using pole-placement, design a gain K so that the closed-loop system has a pair of eigenvalues at $s = -(n_3 + 1) \pm j(n_4 + 1)$, where $j = \sqrt{-1}$.

(8 marks)

- (b) Using the gain K from sub-problem (a) find N such that under the control law $u = -Kx + Nr$, where $r(t)$ is a step input, the closed-loop system can follow the reference signal r with no error.

(8 marks)

- (c) Suppose that you cannot access the state $x(t)$ and can only access the output $y(t)$. Design a Luenberger observer so that the closed-loop system with the output feedback control law $u = -K\xi^o + Nr$, where ξ^o is the observer state, has two closed-loop eigenvalues at $-(n_3 + 4)$ and $-(n_3 + 5)$.

(8 marks)

- (d) Take K from sub-problem (a) and N from sub-problem (b). Can the output feedback control law from sub-problem (c) make the closed-loop system follow a step input r with no error? Justify your answer.

(6 marks)

⁰Please turn over - Question 3 on next page.

QUESTION 3

(30 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Answer the following questions.

- (a) A one-state SISO LTI state-space system has the state equation,

$$\dot{x} = (n_3 + 1)x + (2 + n_6)u.$$

Consider the finite horizon LQR problem with horizon $t_f = 1$ and cost function

$$J_1(x_0, u) = \int_0^1 (x(t)^2 + 5u(t)^2)dt + 10x(1)^2.$$

Compute the solution $P(t)$ of the RDE with terminal condition $P(1) = 10$ as a function of t for $0 \leq t \leq 1$.

(6 marks)

- (b) Consider a SISO LTI state-space system with the state equation

$$\dot{x} = \begin{bmatrix} 0 & n_1 + 1 \\ 0 & n_2 \end{bmatrix} x + \begin{bmatrix} 0 \\ n_3 + 2 \end{bmatrix} u.$$

Let $x = (x_1, x_2)^T$. Suppose that you could access all the state variables. Consider the infinite horizon LQR problem of minimising

$$J_\infty(x_0, u) = \int_0^\infty (25x_1(t)^2 + u(t)^2)dt.$$

- (i) Show that the infinite horizon problem is solvable.

(5 marks)

- (ii) Solve the CARE equation.

(8 marks)

- (iii) Compute the optimal infinite-horizon LQR cost when starting from $x_0 = \begin{bmatrix} 1 & -5 \end{bmatrix}^T$.

(3 marks)

- (c) Consider the nonlinear controlled ODE:

$$\begin{aligned} \dot{x}_1 &= -5 - 2x_1 + x_2 - (x_1 + 2)(x_2 - 1) + u, \\ \dot{x}_2 &= 10 + 4x_1 - 2x_2 + 3(x_1 + 2)^2 + 10(x_2 - 1)^2 + 5u. \end{aligned}$$

Let $x = (x_1, x_2)^T$. Design a state feedback gain K such that the control law $u = -K(x - x^*)$ renders the point $x^* = (-2, 1)^T$ asymptotically stable and the closed-loop Jacobian has eigenvalues at $-(n_5 + 1) \pm j(n_6 + 1)$, where $j = \sqrt{-1}$.

(8 marks)

⁰Please turn over - Question 4 on next page.

QUESTION 4

(20 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Answer the following questions:

(a) Let X be the symmetric matrix variable,

$$X = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_1 \end{bmatrix}$$

and consider the LMI,

$$\begin{bmatrix} -n_4 + 10 & n_4 - 9 \\ n_4 - 9 & -n_4 + 10 \end{bmatrix} X \begin{bmatrix} -n_4 + 10 & n_4 - 9 \\ n_4 - 9 & -n_4 + 10 \end{bmatrix} - 400X < 0.$$

(i) Express the LMI in terms of the vector decision variable $x = (x_1, x_2)^T$. (4 marks)

(ii) By sketching, determine whether there exists a region in $\mathbb{R}^2$ where the LMI is feasible. (8 marks)

(b) Let $A \in \mathbb{R}^{n \times m}$ and $Q = Q^T \in \mathbb{R}^{n \times n}$ be given non-zero constant matrices. Consider the following matrix inequalities in the matrix variables $X \in \mathbb{R}^{m \times m}$ and $Y = Y^T \in \mathbb{R}^{n \times n}$,

$$\begin{aligned} Y &> 0 \\ YAX + X^T A^T Y - Y(XY^{-1}X^T - Q)Y &> 0. \end{aligned}$$

By introducing the new variables $Z = Y^{-1}$ and $W = XY^{-1}$ show that the matrix inequalities can be expressed as an LMI. (8 marks)